

ADVANCED WEB TECHNOLOGY

UNIT I

Introduction to HTML

Internet:

An internet (small i) is simply a set networks connected by devices that exchange data with each other in packets. The Internet is the name given to the largest set of interconnected internets in the world. In particular, the method by which this data is exchanged on the Internet is through the TCP/IP protocol suite.

The Internet is not run by any one organization nor operated by any single agency. This makes a definition of the Internet that identifies a single organization who owns it or runs it impossible. In truth, the Internet is run by a vast patchwork of telecommunications organizations, research centers, universities, and private individuals through their mutual cooperation to ensure the exchange of data among their networks in packets using a common TCP/IP protocol.

WWW:

The Web, or World Wide Web, is basically a system of Internet servers that support specially formatted documents. The documents are formatted in a markup language called HTML (HyperText Markup Language) that supports links to other documents, as well as graphics, audio, and video files.

URL:

A URL (Uniform Resource Locator), as the name suggests, provides a way to locate a resource on the web, the hypertext system that operates over the internet. The URL contains the name of the protocol to be used to access the resource and a resource name. The first part of a URL identifies what protocol to use. The second part identifies the IP address or domain name where the resource is located.

Webpage:

A web page or webpage is a document commonly written in HyperText Markup Language (HTML) that is accessible through the Internet or other network using an Internet browser. A web page is accessed by entering a URL address and may contain text, graphics, and hyperlinks to other web pages and files.

Website:

Virtual location on WWW, containing several subject or company related webpages and data files accessible through a browser. Each website has its own unique web address (see uniform resource

locator) which can be reached through an internet connection. The opening page of a website is usually called homepage which contains hyperlinks to other pages on the same or other site(s). A single web server may support multiple websites and a single website may reside on multiple web servers, sometimes thousands of miles apart.

Hypertext and Hypertext Markup Language

The Hypertext Transfer Protocol (HTTP) is the set of rules for exchanging files (text, graphic images, sound, video, and other multimedia files) on the World Wide Web. Relative to the TCP/IP suite of protocols, HTTP is an application protocol.

Any Web server machine contains, in addition to the HTML and other files it can serve, an HTTP daemon, a program that is designed to wait for HTTP requests and handle them when they arrive. Your Web browser is an HTTP client, sending requests to server machines. When the browser user enters file requests by either "opening" a Web file (typing in a URL) or clicking on a hypertext link, the browser builds an HTTP request and sends it to the Internet Protocol address indicated by the URL. The HTTP daemon in the destination server machine receives the request and, after any necessary processing, the requested file is returned.

What is HTML?

Hyper Text Markup Language, it is the heart of each website because every site finally renders HTML into the browser no matter whether it is developed in any technology like Java, ASP.NET, and PHP etc. HTML is not a programming language; it is a markup language. It is very powerful and easy to learn. It does not require any software to start; you can easily start from text editor. That makes it cost effective. Performance of HTML is very high to any other technology.

What is HTML5?

HTML5 is the new standard of HTML. It has enhanced features for programmers as well as for users. Users will not have to use a Flash plug-in for video and audio content. It fully supports CSS3, 2D/3D Graphics, Video and Audio etc. Visual Studio users typically write code in HTML5 when creating web site. It also supports Local SQL Database that makes it more effective.

What is CSS?

CSS stands for Cascading Style Sheet, it is the backend of the html. Presentation of your site mainly depends upon the styles. Styles are used to display HTML elements in well suited format. Styles can define in CSS that saves your time and you can easy control your site from one place. It is also very powerful; you can even control your rendering of styles in particular browser.

What is PHP?

PHP stands for PHP: Hypertext Preprocessor. It is a server side technology like ASP. Unlike HTML, PHP needs server to run. You can freely download this software from the internet. It is open source software; you can easily customize this software according to your requirement. There are lots of plug-in available in the market that makes its development faster. Currently most of the sites are built in PHP rather than others technologies because of shorter time of development. There are a number of technologies that used PHP like Word Press, OS Commerce, Joomla, Magento etc. PHP supports many databases (MySQL, Informix, Oracle, Sybase, Solid, PostgreSQL, Generic ODBC, etc.)

What is MYSQL?

My SQL is an ORACLE product and usually used in PHP and JAVA. It is used to store, update, delete and retrieve data. It is very powerful software and used in big and small application. It is a free software. MySQL is also used in large application including, Google and Facebook. MySQL is very flexible and its performance is very high.

HTML

HTML is the standard markup language for creating Web pages.

- HTML stands for Hyper Text Markup Language
- HTML describes the structure of Web pages using markup
- HTML elements are the building blocks of HTML pages
- HTML elements are represented by tags
- HTML tags label pieces of content such as "heading", "paragraph", "table", and so on
- Browsers do not display the HTML tags, but use them to render the content of the page

A Simple HTML Document

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

Example Explained

- The `<!DOCTYPE html>` declaration defines this document to be HTML5
- The `<html>` element is the root element of an HTML page
- The `<head>` element contains meta information about the document
- The `<title>` element specifies a title for the document
- The `<body>` element contains the visible page content
- The `<h1>` element defines a large heading
- The `<p>` element defines a paragraph

HTML Tags

HTML tags are element names surrounded by angle brackets:

```
<tagname>content goes here...</tagname>
```

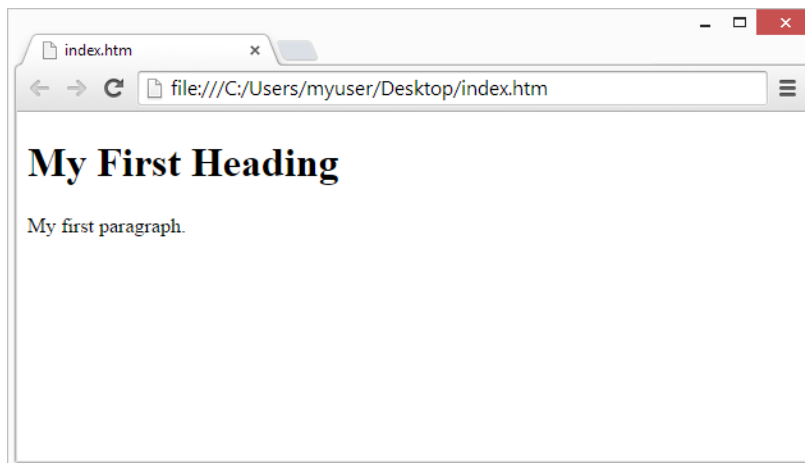
- HTML tags normally come **in pairs** like `<p>` and `</p>`
- The first tag in a pair is the **start tag**, the second tag is the **end tag**
- The end tag is written like the start tag, but with a **forward slash** inserted before the tag name

Tip: The start tag is also called the **opening tag**, and the end tag the **closing tag**.

Web Browsers

The purpose of a web browser (Chrome, IE, Firefox, Safari) is to read HTML documents and display them.

The browser does not display the HTML tags, but uses them to determine how to display the document:



HTML Page Structure

Below is a visualization of an HTML page structure:

```
<html>
<head>
```

```
<title>Page title</title>
</head>
<body>
<h1>This is a heading</h1>
<p>This is a paragraph</p>
<p>This is another paragraph</p>
</body>
</html>
```

Note: Only the content inside the <body> section (the white area above) is displayed in a browser.

The <!DOCTYPE> Declaration

- The <!DOCTYPE> declaration represents the document type, and helps browsers to display web pages correctly.
- It must only appear once, at the top of the page (before any HTML tags).
- The <!DOCTYPE> declaration is not case sensitive.
- The <!DOCTYPE> declaration for HTML is:

<!DOCTYPE html>

HTML Versions

Since the early days of the web, there have been many versions of HTML:

Version	Year
HTML	1991
HTML 2.0	1995
HTML 3.2	1997
HTML 4.01	1999
XHTML	2000
HTML5	2014

HTML Editors

Write HTML Using Notepad or TextEdit

- Web pages can be created and modified by using professional HTML editors.
- However, for learning HTML we recommend a simple text editor like Notepad (PC) or TextEdit (Mac).
- We believe using a simple text editor is a good way to learn HTML.

Follow the four steps below to create your first web page with Notepad or TextEdit.

Step 1: Open Notepad (PC)

- Windows 8 or later:
- Open the Start Screen (the window symbol at the bottom left on your screen). Type Notepad.
- Windows 7 or earlier:
- Open Start > Programs > Accessories > Notepad

Step 1: Open TextEdit (Mac)

- Open Finder > Applications > TextEdit
- Also change some preferences to get the application to save files correctly. In Preferences > Format > choose "Plain Text"
- Then under "Open and Save", check the box that says "Ignore rich text commands in HTML files".
- Then open a new document to place the code.

Step 2: Write Some HTML

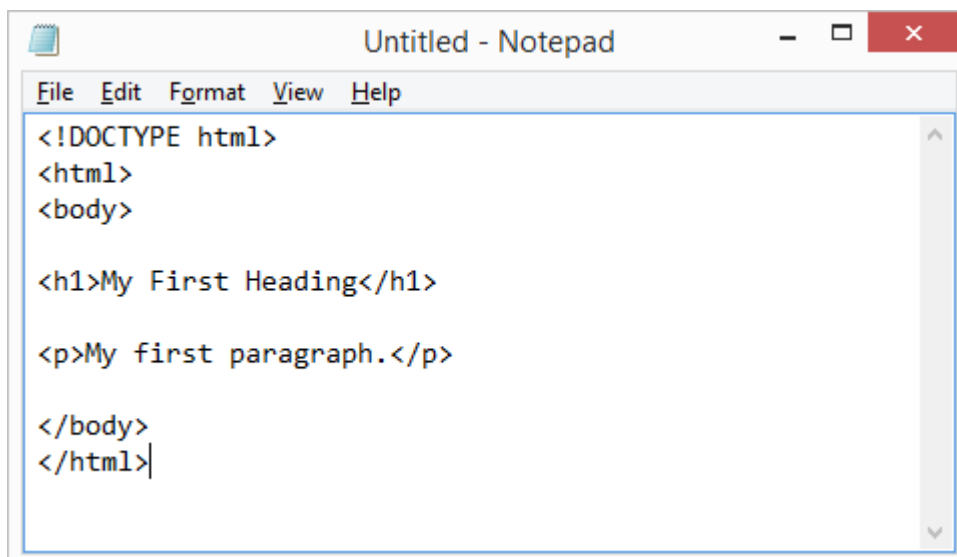
Write or copy some HTML into Notepad.

```
<!DOCTYPE html>
<html>
<body>

<h1>My First Heading</h1>

<p>My first paragraph.</p>

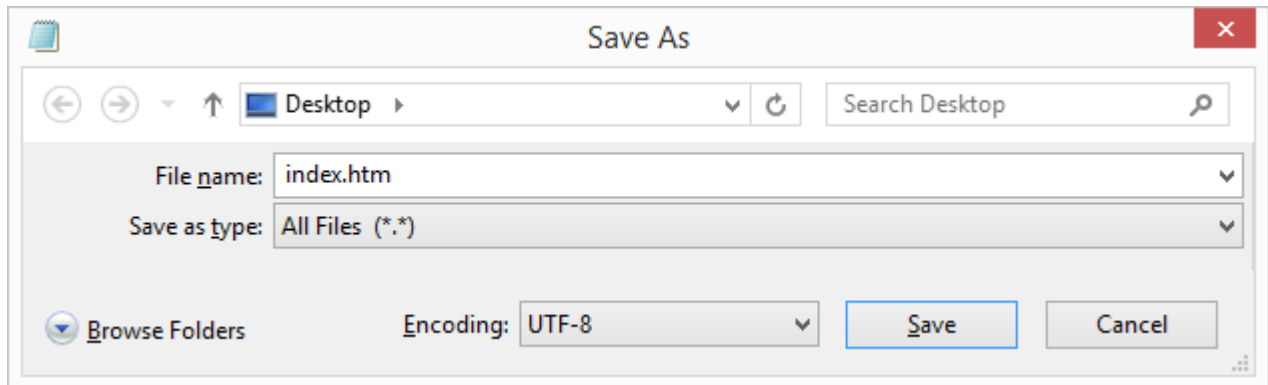
</body>
</html>
```



Step 3: Save the HTML Page

Save the file on your computer. Select **File > Save as** in the Notepad menu.

Name the file "**index.htm**" and set the encoding to **UTF-8** (which is the preferred encoding for HTML files).

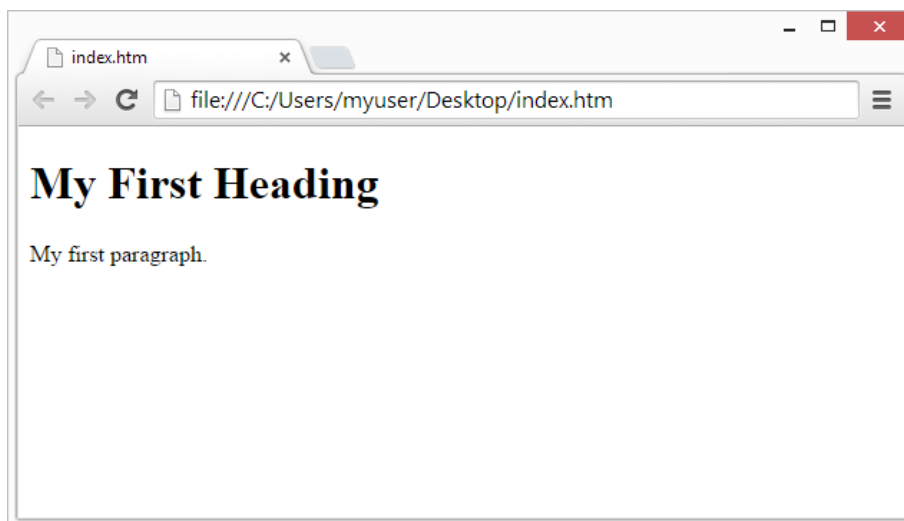


You can use either .htm or .html as file extension. There is no difference, it is up to you.

Step 4: View the HTML Page in Your Browser

Open the saved HTML file in your favorite browser (double click on the file, or right-click - and choose "Open with").

The result will look much like this:



HTML Basic Examples

HTML Documents

- All HTML documents must start with a document type declaration: `<!DOCTYPE html>`.
- The HTML document itself begins with `<html>` and ends with `</html>`.
- The visible part of the HTML document is between `<body>` and `</body>`.

Example

```
<!DOCTYPE html>
<html>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

<Body> Element

HTML body section is a main contain section of web page all contain that will be seen when the user loads the webpage.

HTML body section supported all the contains such as text, hyperlinks, images, Special Character, lists, tables, frames, forms etc.

It's most powerful section and important section to display web page.

Basic Body Section Tags:

Tag	Description
<code><a></code>	Defines the internal link, external link and Subdirectory link.
<code>
</code>	Defines the single line break.
<code><code></code>	Defines the computer code on text base.
<code><div></code>	Defines the division section in a document.
<code><form></code>	Defines the HTML form for use in input documents
<code><frame></code>	Defines the many individual part
<code><hr /></code>	Defines the horizontal line.
<code><p></code>	Defines the paragraph in web documents.
<code><pre></code>	Defines the preformatted text.
<code></code>	Defines a section in a web document.
<code><table></code>	Defines the table.
<code><textarea></code>	Defines the multiline text.

HTML Headings

Headings are important in HTML documents.

Headings are defined with the `<h1>` to `<h6>` tags.

`<h1>` defines the most important heading. `<h6>` defines the least important heading.

Example

```
<h1>This is heading 1</h1>
<h2>This is heading 2</h2>
<h3>This is heading 3</h3>
<h4>This is heading 4</h4>
<h5>This is heading 5</h5>
<h6>This is heading 6</h6>
```

Note: Browsers automatically add some white space (a margin) before and after a heading.

Headings Are Important

- Search engines use the headings to index the structure and content of your web pages.
- Users skim your pages by its headings. It is important to use headings to show the document structure.
- `<h1>` headings should be used for main headings, followed by `<h2>` headings, then the less important `<h3>`, and so on.

Note: Use HTML headings for headings only. Don't use headings to make text **BIG** or **bold**.

The HTML `<head>` Element

- The HTML `<head>` element has nothing to do with HTML headings.
- The `<head>` element is a container for metadata. HTML metadata is data about the HTML document. Metadata is not displayed.
- The `<head>` element is placed between the `<html>` tag and the `<body>` tag:

Example

```
<!DOCTYPE html>
<html>

  <head>
    <title>My First HTML</title>
    <meta charset="UTF-8">
  </head>

  <body>
    .
```

HTML Paragraphs

HTML paragraphs are defined with the **<p>** tag:

Example

```
<p>This is a paragraph.</p>
<p>This is another paragraph.</p>
```

HTML Links

HTML links are defined with the **<a>** tag:

Example

```
<a href="http://www.w3schools.com">This is a link</a>
```

- The link's destination is specified in the **href attribute**.
- Attributes are used to provide additional information about HTML elements.

HTML Images

- HTML images are defined with the **** tag.
- The source file (src), alternative text (alt), width, and height are provided as attributes:

Example

```

```

HTML Elements

An HTML element usually consists of a **start** tag and **end** tag, with the content inserted in between:

```
<tagname>Content goes here...</tagname>
```

The HTML **element** is everything from the start tag to the end tag:

```
<p>My first paragraph.</p>
```

Start tag	Element content	End tag
<h1>	My First Heading	</h1>

<p>	My first paragraph.	</p>

HTML elements with no content are called empty elements. Empty elements do not have an end tag, such as the
 element (which indicates a line break).

● Nested HTML Elements

- HTML elements can be nested (elements can contain elements).
- All HTML documents consist of nested HTML elements.
- This example contains four HTML elements:

Example

```
<!DOCTYPE html>
<html>
  <body>

    <h1>My First Heading</h1>
    <p>My first paragraph.</p>

  </body>
</html>
```

Example Explained

The <html> element defines the **whole document**.

It has a **start** tag <html> and an **end** tag </html>.

The element **content** is another HTML element (the <body> element).

```
<html>
  <body>

    <h1>My First Heading</h1>
    <p>My first paragraph.</p>

  </body>
</html>
```

- The **<body>** element defines the **document body**.
- It has a **start** tag **<body>** and an **end** tag **</body>**.
- The element **content** is two other HTML elements (**<h1>** and **<p>**).

```
<body>

  <h1>My First Heading</h1>
  <p>My first paragraph.</p>

</body>
```

- The **<h1>** element defines a **heading**.
- It has a **start** tag **<h1>** and an **end** tag **</h1>**.
- The element **content** is: My First Heading.

```
<h1>My First Heading</h1>
```
- The **<p>** element defines a **paragraph**.
- It has a **start** tag **<p>** and an **end** tag **</p>**.
- The element **content** is: My first paragraph.

```
<p>My first paragraph.</p>
```

● Empty HTML Elements

- HTML elements with no content are called empty elements.
- **
** is an empty element without a closing tag (the **
** tag defines a line break).
- Empty elements can be "closed" in the opening tag like this: **
**.
- HTML5 does not require empty elements to be closed. But if you want stricter validation, or if you need to make your document readable by XML parsers, you must close all HTML elements properly.

● HTML Attributes

- HTML tags contain have one or more attributes are support.
- HTML Attributes are added in tag to provide the more additional information about how the tag should be appear or behavior.
- HTML Attributes are always specified in the start tag.
- HTML Attributes consist on name/value pairs like: i.e. name="value" and separated by an equals (=) sign.
- Attribute values always be enclosed in double/single quotes.
- Double quotes are the most common use, but single quotes are also allowed.

● The lang Attribute

- The language of the document can be declared in the **<html>** tag.
- The language is declared with the **lang** attribute.

- Declaring a language is important for accessibility applications (screen readers) and search engines:

```
<!DOCTYPE html>
<html lang="en-US">
<body>
```

...

```
</body>
</html>
```

The first two letters specify the language (en). If there is a dialect, use two more letters (US).

● The title Attribute

Here, a **title** attribute is added to the **<p>** element. The value of the title attribute will be displayed as a tooltip when you mouse over the paragraph:

Example

```
<p title="I'm a tooltip">
This is a paragraph.
</p>
```

● The href Attribute

HTML Internal Link is linked within the same web page. This link can be absolute path or relative path.

HTML internal link name is followed by hash sign(#). HTML **<a>** tag is used for anchor point name, which is referred to an internal link into a same page.

When click on this anchor link, it's referred link automatically scrolls and displays on browser.

HTML links are defined with the **<a>** tag. The link address is specified in the **href** attribute:

Example

```
<a href="http://www.w3schools.com">This is a link</a>
<a href="html.php">HTML</a> <br />
<br />
<a href="css.php">CSS</a> <br />
<br />
<a href="javascript.php">Java Script</a> <br />
```

 Element

- HTML images are defined with the **** tag.

- The filename of the source (**src**), and the size of the image (**width** and **height**) are all provided as **attributes**:

Example

```

```

The image size is specified in pixels: width="104" means 104 screen pixels wide.

● The alt Attribute

- The **alt** attribute specifies an alternative text to be used, when an image cannot be displayed.
- The value of the attribute can be read by screen readers. This way, someone "listening" to the webpage, e.g. a blind person, can "hear" the element.

Example

```

```

● Formatting Tags

HTML formatting tags use for formatting a text style. It will become necessary to make minor changes to the formatting of those elements.

HTML Tags like for formatting output **bold** text.

Tag	Example	Results
	Bold Text	An example of Bold Text
<big>	<big>Big Text</big>	An example of Big Text
<center>	<center>Center Text</center>	An example of Center Text
	Emphasized Text	An example of <i>Emphasized Text</i>
<i>	<i>Italic Text</i>	An example of <i>Italic Text</i>
<small>	Small Text	An example of Small Text
	Strong Text	An example of Strong Text
<sub>	_{Subscript Text}	An example of Subscript Text
<sup>	^{Superscript Text}	An example of Superscript Text
	Delete Text	An example of Delete Text
<s>	<s>Strike Text</s>	An example of Strike Text
<strike>	<strike>Strike Text</strike>	An example of Strike Text
<u>	<u>Underline Text</u>	An example of <u>Underline Text</u>
<tt>	<tt>Teletype Text</tt>	An example of Teletype Text

<blockquote>	<blockquote>Long Quotation</blockquote> <q>Short Quotation Text</q>	An example of Long Quotation Text An example of Short Quotation Text
--------------	---	---

If you use a word processor, you must be familiar with the ability to make text bold, italicized, or underlined; these are just three of the ten options available to indicate how text can appear in HTML and XHTML.

● **Bold Text**

Anything that appears within **...** element, is displayed in bold as shown below:

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Bold Text Example</title>
</head>
<body>
<p>The following word uses a <b>bold</b> typeface.</p>
</body>
</html>
```

This will produce following result:

The following word uses a **bold** typeface.

● **Italic Text**

Anything that appears within **<i>...</i>** element is displayed in italicized as shown below:

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Italic Text Example</title>
</head>
<body>
<p>The following word uses a <i>italicized</i> typeface.</p>
</body>
</html>
```

This will produce following result:

The following word uses a *italicized* typeface.

● Underlined Text

Anything that appears within `<u>...</u>` element, is displayed with underline as shown below:

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Underlined Text Example</title>
</head>
<body>
<p>The following word uses a <u>underlined</u> typeface.</p>
</body>
</html>
```

This will produce following result:

The following word uses a underlined typeface.

● Strike Text

Anything that appears within `<strike>...</strike>` element is displayed with strikethrough, which is a thin line through the text as shown below:

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Strike Text Example</title>
</head>
<body>
<p>The following word uses a <strike>strikethrough</strike> typeface.</p>
</body>
</html>
```

This will produce following result:

The following word uses a ~~strikethrough~~ typeface.

● Monospaced Font

The content of a `<tt>...</tt>` element is written in monospaced font. Most of the fonts are known as variable-width fonts because different letters are of different widths (for example, the letter 'm' is wider than the letter 'i'). In a monospaced font, however, each letter has the same width.

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Monospaced Font Example</title>
```



```
</head>
<body>
<p>The following word uses a <tt>monospaced</tt> typeface.</p>
</body>
</html>
```

This will produce following result:

The following word uses a monospaced typeface.

● Superscript Text

The content of a **^{...}** element is written in superscript; the font size used is the same size as the characters surrounding it but is displayed half a character's height above the other characters.

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Superscript Text Example</title>
</head>
<body>
<p>The following word uses a <sup>superscript</sup> typeface.</p>
</body>
</html>
```

This will produce following result:

The following word uses a ^{superscript} typeface.

● Subscript Text

The content of a **_{...}** element is written in subscript; the font size used is the same as the characters surrounding it, but is displayed half a character's height beneath the other characters.

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Subscript Text Example</title>
</head>
<body>
<p>The following word uses a <sub>subscript</sub> typeface.</p>
</body>
</html>
```

This will produce following result:

The following word uses a _{subscript} typeface.

● **Inserted Text**

Anything that appears within `<ins>...</ins>` element is displayed as inserted text.

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Inserted Text Example</title>
</head>
<body>
<p>I want to drink <del>cola</del> <ins>wine</ins></p>
</body>
</html>
```

This will produce following result:

I want to drink ~~cola~~ wine

● **Deleted Text**

Anything that appears within `<del>...</del>` element, is displayed as deleted text.

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Deleted Text Example</title>
</head>
<body>
<p>I want to drink <del>cola</del> <ins>wine</ins></p>
</body>
</html>
```

This will produce following result:

I want to drink ~~cola~~ wine

● **Larger Text**

The content of the `<big>...</big>` element is displayed one font size larger than the rest of the text surrounding it as shown below:

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Larger Text Example</title>
```

```
</head>
<body>
<p>The following word uses a <big>big</big> typeface.</p>
</body>
</html>
```

This will produce following result:

The following word uses a big typeface.

● **Smaller Text**

The content of the **<small>...</small>** element is displayed one font size smaller than the rest of the text surrounding it as shown below:

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Smaller Text Example</title>
</head>
<body>
<p>The following word uses a <small>small</small> typeface.</p>
</body>
</html>
```

This will produce following result:

The following word uses a small typeface.

● **Grouping Content**

The **<div>** and **** elements allow you to group together several elements to create sections or subsections of a page.

For example, you might want to put all of the footnotes on a page within a **<div>** element to indicate that all of the elements within that **<div>** element relate to the footnotes. You might then attach a style to this **<div>** element so that they appear using a special set of style rules.

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Div Tag Example</title>
</head>
<body>
<div id="menu" align="middle" >
<a href="/index.htm">HOME</a> |
<a href="/about/contact_us.htm">CONTACT</a> |
```

```
<a href="/about/index.htm">ABOUT</a>
</div>
```

```
<div id="content" align="left" bgcolor="white">
<h5>Content Articles</h5>
<p>Actual content goes here.....</p>
</div>
</body>
</html>
```

This will produce following result:

[HOME](#) | [CONTACT](#) | [ABOUT](#)

Content Articles

Actual content goes here.....

The `<span>` element, on the other hand, can be used to group inline elements only. So, if you have a part of a sentence or paragraph which you want to group together, you could use the `<span>` element as follows

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Span Tag Example</title>
</head>
<body>
<p>This is the example of <span style="color:green">span tag</span> and the <span
style="color:red">div tag</span> alongwith CSS</p>
</body>
</html>
```

Single or Double Quotes?

- Double quotes around attribute values are the most common in HTML, but single quotes can also be used.
- In some situations, when the attribute value itself contains double quotes, it is necessary to use single quotes:

```
<p title='John "ShotGun" Nelson'>
```

Or vice versa:

```
<p title="John 'ShotGun' Nelson">
```

HTML Horizontal Rules

- The `<hr>` tag defines a thematic break in an HTML page, and is most often displayed as a horizontal rule.
- The `<hr>` element is used to separate content (or define a change) in an HTML page:

Example

```
<h1>This is heading 1</h1>
<p>This is some text.</p>
<hr>
<h2>This is heading 2</h2>
<p>This is some other text.</p>
<hr>
```

HTML Display

- You cannot be sure how HTML will be displayed.
- Large or small screens, and resized windows will create different results.
- With HTML, you cannot change the output by adding extra spaces or extra lines in your HTML code.
- The browser will remove any extra spaces and extra lines when the page is displayed:

Example

```
<p>
This paragraph
contains a lot of lines
in the source code,
but the browser
ignores it.
</p>

<p>
This paragraph
contains      a lot of spaces
in the source    code,
but the      browser
ignores it.
</p>
```

Don't Forget the End Tag

Most browsers will display HTML correctly even if you forget the end tag:

Example

```
<p>This is a paragraph.
<p>This is another paragraph.
```

HTML Line Breaks

The HTML `<br>` element defines a **line break**.

Use `<br>` if you want a line break (a new line) without starting a new paragraph:

Example

```
<p>This is<br>a paragraph<br>with line breaks.</p>
```

The `<br>` tag is an empty tag, which means that it has no end tag.

The Poem Problem

This poem will display on a single line:

Example

```
<p>
  My Bonnie lies over the ocean.

  My Bonnie lies over the sea.

  My Bonnie lies over the ocean.

  Oh, bring back my Bonnie to me.
</p>
```

The HTML `<pre>` Element

The HTML `<pre>` element defines preformatted text.

The text inside a `<pre>` element is displayed in a fixed-width font (usually Courier), and it preserves both spaces and line breaks:

Example

```
<pre>
  My Bonnie lies over the ocean.

  My Bonnie lies over the sea.

  My Bonnie lies over the ocean.

  Oh, bring back my Bonnie to me.
</pre>
```

HTML Styles

I am red

I am blue

The HTML Style Attribute

- Setting the style of an HTML element, can be done with the **style attribute**.
- The HTML style attribute has the following **syntax**:
- `<tagname style="property:value;">`
- The **property** is a CSS property. The **value** is a CSS value.

HTML Background Color

The **background-color** property defines the background color for an HTML element.

This example sets the background color for a page to powderblue:

Example

```
<body style="background-color:powderblue;">
```

```
<h1>This is a heading</h1>
```

```
<p>This is a paragraph.</p>
```

```
</body>
```

HTML Text Color

The **color** property defines the text color for an HTML element:

Example

```
<h1 style="color:blue;">This is a heading</h1>
```

```
<p style="color:red;">This is a paragraph.</p>
```

HTML Fonts

The **font-family** property defines the font to be used for an HTML element:

Example

```
<h1 style="font-family:verdana;">This is a heading</h1>
```

```
<p style="font-family:courier;">This is a paragraph.</p>
```

HTML Text Size

The **font-size** property defines the text size for an HTML element:

Example

```
<h1 style="font-size:300%;">This is a heading</h1>
```

```
<p style="font-size:160%;">This is a paragraph.</p>
```

HTML Text Alignment

The **text-align** property defines the horizontal text alignment for an HTML element:

Example

```
<h1 style="text-align:center;">Centered Heading</h1>
<p style="text-align:center;">Centered paragraph.</p>
```

HTML Comment Tags

- You can add comments to your HTML source by using the following syntax:
- `<!-- Write your comments here -->`
- Notice that there is an exclamation point (!) in the opening tag, but not in the closing tag.
- **Note:** Comments are not displayed by the browser, but they can help document your HTML source code.
- With comments you can place notifications and reminders in your HTML:

Example

```
<!-- This is a comment -->

<p>This is a paragraph.</p>

<!-- Remember to add more information here -->
```

Comments are also great for debugging HTML, because you can comment out HTML lines of code, one at a time, to search for errors:

Example

```
<!-- Do not display this at the moment

-->
```

HTML Lists

An Unordered List:

- Item
- Item
- Item
- Item

An Ordered List:

1. First item
2. Second item
3. Third item
4. Fourth item

Unordered HTML List

An unordered list starts with the `<ul>` tag. Each list item starts with the `<li>` tag.

The list items will be marked with bullets (small black circles) by default:

```
<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

Output:

An unordered HTML list

- Coffee
- Tea
- Milk

Unordered HTML List - Choose List Item Marker

The CSS **list-style-type** property is used to define the style of the list item marker:

Value	Description
disc	Sets the list item marker to a bullet (default)
circle	Sets the list item marker to a circle
square	Sets the list item marker to a square
none	The list items will not be marked

```
<ul style="list-style-type:disc">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

Ordered HTML List

An ordered list starts with the `<ol>` tag. Each list item starts with the `<li>` tag.

The list items will be marked with numbers by default:

Example

```
<ol>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

Ordered HTML List - The Type Attribute

The **type** attribute of the tag, defines the type of the list item marker:

Type	Description
type="1"	The list items will be numbered with numbers (default)
type="A"	The list items will be numbered with uppercase letters
type="a"	The list items will be numbered with lowercase letters
type="I"	The list items will be numbered with uppercase roman numbers
type="i"	The list items will be numbered with lowercase roman numbers

```
<ol type="1">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

Uppercase Letters:

```
<ol type="A">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

Lowercase Letters:

```
<ol type="a">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

Uppercase Roman Numbers:

```
<ol type="I">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

Lowercase Roman Numbers:

```
<ol type="i">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

HTML Marquee Tag

- It is used to create a scrolling text or scrolling image from left to right, right to left, top to bottom, bottom to top. There is no limit. It's user define choice.
- <marquee> tag is a Container tag to create scrolling text.
- <marquee> tag support following some attributes. This attributes use to add more special effect and easy control.

Marquee Attributes

Attributes	Values	Description
behavior	"slide" "scroll" "alternate"	Start completely and stop as soon as text touches the margin. Start completely and off one side. Text bounce as soon as touch both side margin.
bgcolor	"Color_Code"	Specifies the background color.
direction	"left" "right" "up" "down"	Left to Right Right to Left Bottom to Top Top to Bottom
width	"size_px"	Specifies width in marquee.
height	"size_px"	Specifies height in marquee.
loop	"2" "infinite"	Loop Continues in limited times. Loop Continues in Infinite.
scrollamount	"3"	Specifies Speed to scroll on the text.
scrolldelay	"3"	Specifies time wait long before each marquee.

hspace	"size_px"	Specifies left or right margin for outside marquee.
vspace	"size_px"	Specifies top or bottom margin for outside marquee.

Example:

```
<marquee behavior="scroll" direction="left"
onmousedown="this.stop();" onmouseup="this.start();">
Click Here and hold the mouse marquee stop
</marquee>
<p>
<a
href="http://www.way2tutorial.com/html/html_marquee_start_stop.ph
p">Marquee Start/Stop</a>
</p>
```

<Table> Element

- The best way to split a page into different sections is to use HTML <table> Tag.
- HTML Table section is top of (start with) <table> tag, and closed it right down at the </table> tag. Everything in between those two tags is inside the table, as you can see in following screenshot.
- <table> tag inside you need two more tag first is <tr> tag, which stands for table row. It is closed with the </tr> tag. And another one is <td> tag, which stands for table data.
- Every time you add a <tr> tag, when table will gain an extra row. In the table you just made, these is only one row section, so these is only one row.
- HTML <td> tag stands for table data (also say table column), and it places one cell in your table row.
- HTML table you also merge two or more column or two or more row merge using respectively **colspan** or **rowspan** attributes.

```
<!DOCTYPE html>
<html>
<head>
<title>HTML Table Header</title>
</head>
<body>
<table border="1">
<tr>
<th>Name</th>
<th>Salary</th>
</tr>
<tr>
<td>Ramesh Raman</td>
<td>5000</td>
</tr>
```

```

<tr>
<td>Shabbir Hussein</td>
<td>7000</td>
</tr>
</table>
</body>
</html>

```

This will produce following result:

Name	Salary
Ramesh Raman	5000
Shabbir Hussein	7000

<table> Attributes

Attributes	Value	Description
align	Left,Right,Center	Declared your alignment side.
width	"Size_px"	Specify the Width Size of the Table.
height	"Size_px"	Specify the Height Size of the Table.
bgcolor	"purple"	Specify the Background Color.
background	"Specified_URL"	Specify the Background Image open for URL file.
border	"Size_px"	Specify the size of border thickness.
bordercolor	"Yellow"	Specify the color of border.
cellspacing	"Size_px"	Specify the space between two cells.
cellpadding	"Size_px"	Specify the space between cell boundary and text.

<th> Attributes

Attributes	Value	Description
bgcolor	"purple"	Specify the Background Color.
colspan	"Column_N"	Specify the span of there column.
rowspan	"Row_N"	Specify the span of there Row.

<tr> Attributes

Attributes	Value	Description
bicolor	"purple"	Specify the Background Color.

A Table with No Cell Padding or Spacing with a Border of 1

```
<TABLE BORDER=1>
<TR>
<TD>Red</TD>
<TD>Green</TD>
<TD>Blue</TD>
</TR>
<TR>
<TD>Orange</TD>
<TD>Yellow</TD>
<TD>Purple</TD>
</TR>
</TABLE>
```

Output:

Red	Green	Blue
Orange	Yellow	Purple

A Table with a Cell Spacing of 10 with a Border of 1

```

<TABLE BORDER=1
CELLSPACING=10>
<TR>
<TD>Red</TD>
<TD>Green</TD>
<TD>Blue</TD>
</TR>
<TR>
<TD>Orange</TD>
<TD>Yellow</TD>
<TD>Purple</TD>
</TR>
</TABLE>

```

Output:

Red	Green	Blue
Orange	Yellow	Purple

An HTML table with a table cell that spans two rows:

```

<table>
<tr>
<th>Month</th>
<th>Savings</th>
<th>Savings for holiday!</th>
</tr>
<tr>
<td>January</td>
<td>$100</td>
<td rowspan="2">$50</td>
</tr>
<tr>
<td>February</td>
<td>$80</td>
</tr>
</table>

```

Output:

Month	Savings	Savings for holiday!
January	\$100	\$50
February	\$80	

An HTML table with a table cell that spans two columns:

```
<table>
  <tr>
    <th>Month</th>
    <th>Savings</th>
  </tr>
  <tr>
    <td>January</td>
    <td>$100</td>
  </tr>
  <tr>
    <td>February</td>
    <td>$80</td>
  </tr>
  <tr>
    <td colspan="2">Sum: $180</td>
  </tr>
</table>
```

Output:

Month	Savings
January	\$100
February	\$80
Sum: \$180	

HTML Forms

The <form> Element

The HTML **<form>** element defines a form that is used to collect user input:

```
<form>
  .
  form elements
  .
</form>
```

- An HTML form contains **form elements**.

- Form elements are different types of input elements, like text fields, checkboxes, radio buttons, submit buttons, and more.

The <input> Element

- The <input> element is the most important form element.
- The <input> element can be displayed in several ways, depending on the **type** attribute.

Here are some examples:

Type	Description
<input type="text">	Defines a one-line text input field
<input type="radio">	Defines a radio button (for selecting one of many choices)
<input type="submit">	Defines a submit button (for submitting the form)

Text Input

<input type="text"> defines a one-line input field for **text input**:

Example

```
<form>
  First name:<br>
  <input type="text" name="firstname"><br>
  Last name:<br>
  <input type="text" name="lastname">
</form>
```

This is how it will look like in a browser:

First name:

Last name:

Note: The form itself is not visible. Also note that the default width of a text field is 20 characters.

Radio Button Input

<input type="radio"> defines a **radio button**.

Radio buttons let a user select ONE of a limited number of choices:

Example

```
<form>
  <input type="radio" name="gender" value="male" checked> Male<br>
  <input type="radio" name="gender" value="female"> Female<br>
  <input type="radio" name="gender" value="other"> Other
</form>
```

This is how the HTML code above will be displayed in a browser:

- ☒ Male
☐ Female
☐ Other

The Submit Button

- **<input type="submit">** defines a button for **submitting** the form data to a **form-handler**.
- The form-handler is typically a server page with a script for processing input data.
- The form-handler is specified in the form's **action** attribute:

Example

```
<form action="action_page.php">
  First name:<br>
  <input type="text" name="firstname" value="Mickey"><br>
  Last name:<br>
  <input type="text" name="lastname" value="Mouse"><br><br>
  <input type="submit" value="Submit">
</form>
```

This is how the HTML code above will be displayed in a browser:

First name:

Last name:

The Action Attribute

- The **action** attribute defines the action to be performed when the form is submitted.
- Normally, the form data is sent to a web page on the server when the user clicks on the submit button.
- In the example above, the form data is sent to a page on the server called "action_page.php". This page contains a server-side script that handles the form data:

```
<form action="action_page.php">
```

If the action attribute is omitted, the action is set to the current page.

The Method Attribute

The **method** attribute specifies the HTTP method (**GET** or **POST**) to be used when submitting the form data:

```
<form action="action_page.php" method="get">
```

or:

```
<form action="action_page.php" method="post">
```

When to Use GET?

- The default method when submitting form data is GET.
- However, when GET is used, the submitted form data will be **visible in the page address field**:
- `action_page.php?firstname=Mickey&lastname=Mouse`
- **Note:** GET must NOT be used when sending sensitive information! GET is best suited for short, non-sensitive, amounts of data, because it has size limitations too.

When to Use POST?

- Always use POST if the form data contains sensitive or personal information. The POST method does not display the submitted form data in the page address field.
- POST has no size limitations, and can be used to send large amounts of data.

The Name Attribute

- Each input field must have a **name** attribute to be submitted.
- If the name attribute is omitted, the data of that input field will not be sent at all.
- This example will only submit the "Last name" input field:

Example

```
<form action="action_page.php">  
  First name:<br>  
  <input type="text" value="Mickey"><br>  
  Last name:<br>  
  <input type="text" name="lastname" value="Mouse"><br><br>  
  <input type="submit" value="Submit">  
</form>
```

Grouping Form Data with <fieldset>

- The **<fieldset>** element is used to group related data in a form.

- The **<legend>** element defines a caption for the **<fieldset>** element.

Example

```
<form action="action_page.php">
  <fieldset>
    <legend>Personal information:</legend>
    First name:<br>
    <input type="text" name="firstname" value="Mickey"><br>
    Last name:<br>
    <input type="text" name="lastname" value="Mouse"><br><br>
    <input type="submit" value="Submit">
  </fieldset>
</form>
```

The **<select>** Element

The **<select>** element defines a **drop-down list**:

Example

```
<select name="cars">
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="fiat">Fiat</option>
  <option value="audi">Audi</option>
</select>
```

- The **<option>** elements defines an option that can be selected.
- By default, the first item in the drop-down list is selected.
- To define a pre-selected option, add the **selected** attribute to the option:

Example

```
<option value="fiat" selected>Fiat</option>
```

The **<textarea>** Element

The **<textarea>** element defines a multi-line input field (**a text area**):

Example

```
<textarea name="message" rows="10" cols="30">
  The cat was playing in the garden.
</textarea>
```

- The **rows** attribute specifies the visible number of lines in a text area.
- The **cols** attribute specifies the visible width of a text area.
- This is how the HTML code above will be displayed in a browser:



The <button> Element

The <button> element defines a clickable **button**:

Example

```
<button type="button" onclick="alert('Hello World!')">Click Me!</button>
```

This is how the HTML code above will be displayed in a browser:

Click Me!

Input Type Password

<input type="password"> defines a **password field**:

Example

```
<form>  
  User name:<br>  
  <input type="text" name="username"><br>  
  User password:<br>  
  <input type="password" name="psw">  
</form>
```

This is how the HTML code above will be displayed in a browser:

User name:

User password:

The characters in a password field are masked (shown as asterisks or circles).

Example

```
<form action="action_page.php">  
  First name:<br>  
  <input type="text" name="firstname" value="Mickey"><br>  
  Last name:<br>  
  <input type="text" name="lastname" value="Mouse"><br><br>
```

```
<input type="submit">  
</form>
```

Input Type Reset

<input type="reset"> defines a **reset button** that will reset all form values to their default values:

Example

```
<form action="action_page.php">  
  First name:<br>  
  <input type="text" name="firstname" value="Mickey"><br>  
  Last name:<br>  
  <input type="text" name="lastname" value="Mouse"><br><br>  
  <input type="submit" value="Submit">  
  <input type="reset">  
</form>
```

This is how the HTML code above will be displayed in a browser:

First name:

Last name:

If you change the input values and then click the "Reset" button, the form-data will be reset to the default values.

Input Type Checkbox

<input type="checkbox"> defines a **checkbox**.

Checkboxes let a user select ZERO or MORE options of a limited number of choices.

Example

```
<form>  
  <input type="checkbox" name="vehicle1" value="Bike"> I have a  
  bike<br>  
  <input type="checkbox" name="vehicle2" value="Car"> I have a car  
</form>
```

This is how the HTML code above will be displayed in a browser:

- ☐ I have a bike
- ☐ I have a car

Input Type Button

`<input type="button">` defines a **button**:

Example

```
<input type="button" onclick="alert('Hello World!')" value="Click Me!">
```

<DIV> Tag

The HTML `<div>` tag is used for defining a section of your document. With the div tag, you can group large sections of HTML elements together and format them with CSS.

The difference between the div tag and the span tag is that the div tag is used with block-level elements while the span tag is used with inline elements.

Example:

```
<div style="color:#0000FF">
  <h3>This is a heading</h3>
  <p>This is a paragraph.</p>
</div>
```

<FRAME> Tag

HTML frames are used to divide your browser window into multiple sections where each section can load a separate HTML document. A collection of frames in the browser window is known as a frameset. The window is divided into frames in a similar way the tables are organized: into rows and columns.

Disadvantages of Frames

There are few drawbacks with using frames, so it's never recommended to use frames in your webpages:

- Some smaller devices cannot cope with frames often because their screen is not big enough to be divided up.
- Sometimes your page will be displayed differently on different computers due to different screen resolution.
- The browser's *back button* might not work as the user hopes.
- There are still few browsers that do not support frame technology.

Creating Frames

To use frames on a page we use `<frameset>` tag instead of `<body>` tag. The `<frameset>` tag defines how to divide the window into frames. The **rows** attribute of `<frameset>` tag defines horizontal frames and **cols** attribute defines vertical frames. Each frame is indicated by `<frame>` tag and it defines which HTML document shall open into the frame.

Example

Following is the example to create three horizontal frames:

```
<!DOCTYPE html>
<html>
<head>
<title>HTML Frames</title>
</head>
<frameset rows="10%,80%,10%">
  <frame name="top" src="/html/top_frame.htm" />
  <frame name="main" src="/html/main_frame.htm" />
  <frame name="bottom" src="/html/bottom_frame.htm" />
</frameset>
<body>
  Your browser does not support frames.
</body>
</frameset>
</html>
```

This will produce following result:



Example

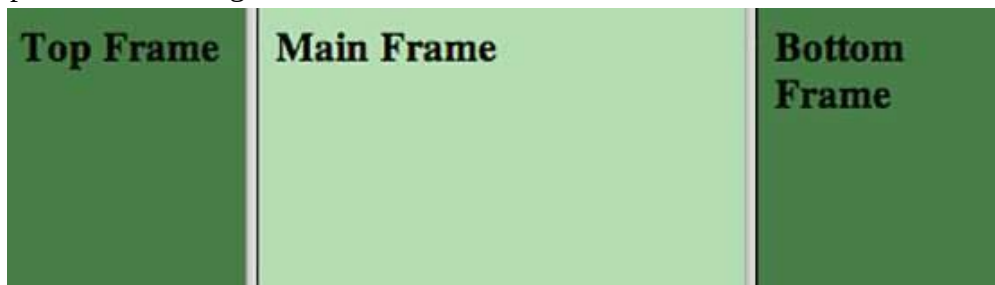
Let's put above example as follows, here we replaced rows attribute by cols and changed their width. This will create all the three frames vertically:

```
<!DOCTYPE html>
<html>
<head>
<title>HTML Frames</title>
</head>
<frameset cols="25%,50%,25%">
  <frame name="left" src="/html/top_frame.htm" />
  <frame name="center" src="/html/main_frame.htm" />
  <frame name="right" src="/html/bottom_frame.htm" />
</frameset>
```



```
<body>
  Your browser does not support frames.
</body>
</noframes>
</frameset>
</html>
```

This will produce following result:



What is XHTML?

- XHTML stands for EXtensible HyperText Markup Language
- XHTML is a stricter, more XML-based version of HTML
- XHTML is HTML defined as an XML application
- XHTML is supported by all major browsers

Why XHTML?

XML is a markup language where all documents must be marked up correctly (be "well-formed"). It was developed to make HTML more extensible and flexible to work with other data formats (such as XML). In addition, browsers ignore errors in HTML pages, and try to display the website even if it has some errors in the markup. So XHTML comes with a much stricter error handling.

The Most Important Differences from HTML

- <!DOCTYPE> is mandatory
- The xmlns attribute in <html> is mandatory
- <html>, <head>, <title>, and <body> are mandatory
- Elements must always be properly nested
- Elements must always be closed
- Elements must always be in lowercase
- Attribute names must always be in lowercase
- Attribute values must always be quoted
- Attribute minimization is forbidden

Meta Tag

The <meta> tag defines metadata about an HTML document. Metadata is data (information) about data. <meta> tags always go inside the <head> element, and are typically used to specify character set, page description, keywords, author of the document, and viewport settings. It will not be displayed on the page, but is machine parsable. It is used by browsers (how to display content or reload page), search engines (keywords), and other web services. There is a method to let web designers take control over the viewport (the user's visible area of a web page), through the <meta> tag.

<frame> and <frameset>

HTML <frame> tag define the particular area within an HTML file where another HTML web page can be displayed.

A <frame> tag is used with <frameset>, and it divides a webpage into multiple sections or frames, and each frame can contain different web pages.

```
<frameset cols="25%,50%,25%">
  <frame src="frame1.html" >
  <frame src="frame2.html">
  <frame src="frame3.html">
</frameset>
```

General Differences between HTML5 and HTML4

HTML4 has been a standard web development for more than 10 years. HTML4 is approved and ratified as a standard language for browsers by the World Wide Web Consortium (W3C), an organization which specifies and approves standards for web technologies.

Following are some of the major characteristics that distinguish HTML5 from HTML4. **1.**

Simplified and Clear Syntax

The syntax in HTML5 is extremely clear and simple as compared to HTML4. One example of this is the DOCTYPE element. In HTML4 the DOCTYPE declaration was too messy and lengthy and used to refer an external source. However in HTML5 DOCTYPE element has been made extremely simple. For instance a mere `<!DOCTYPE html>` is enough to specify the document type.

2. Multimedia Elements

HTML5 contains built in support for integrated multimedia files into web page via video and audio tags. Previously, in HTML4, the multimedia content was integrated in web pages via third party plugins such as Silverlight and flash.

3. Accessing User Geographical location

Previously in HTML4, it was an extremely cumbersome task to get the geographical locations of the visitors visiting the site. It was even difficult when the website was accessed through mobile devices. On the other hand, in HTML5 is extremely easy to get the user location. HTML5's JS GeoLocation can be leveraged to identify the location of the user accessing the website.

4. Client Side storage

In HTML4, in order to store important data on client side, browser's cache was used. However, that cache is limited and doesn't support relational storage mechanism. In HTML5, this issue has been addressed via Web SQL database and application cache that can be access via HTML5's JavaScript interface.

5. Client Server Communication

In HTML4 the communication between the client and server was done through streaming and long polling, since there are no web sockets available in HTML4. On the contrary, HTML5 contains web sockets that allow full duplex communication between clients and servers. **6.**

JavaScript Threading Mechanism

In HTML4, JavaScript and the browser interface with which user interacts, run in the same thread which affects performance. HTML5 contains JS Web Worker API which allows JavaScript and Browser interface to run in separate threads.

7. Browser Compatibility

HTML4 is an established standard for developing browser applications and has been in use for more than 10 years. For this reason, HTML4 is compatible with almost all web-browsers. On the other hand, HTML5 is still in the process of evolution and the currently available tags are being modified and also new tags are being added. Therefore, HTML5 lags behind HTML4 in terms of compatibility with the browsers.